Department of Computer Engineering
Academic Term: July-November 2022

Class : B.*E. Computer*
Semester : VII

Subject Name :*ML*
Subject Code : *CSC701*

Practical No:	7
Title.	**To Study and implement Ensemble Techniques**
Date of Performance.	19/09/2025
Roll No:	9913
Name of the Student:	Mark Lopes

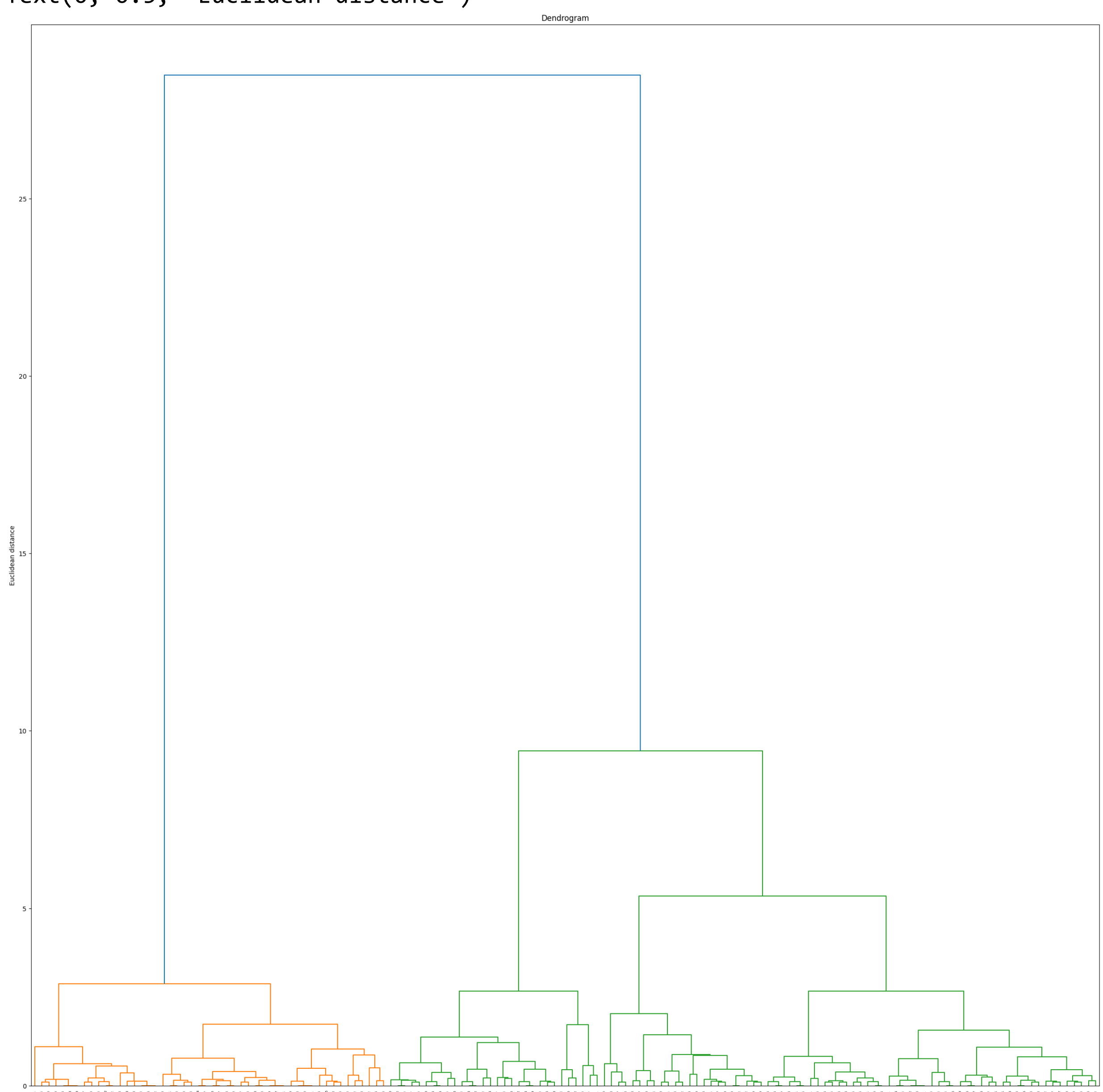
Evaluation

Indicator	**Very Poor**	**Poor**	**Average**	**Good**	**Excellent**
Timeline (2)	More than three sessions late (0)	More than two sessions late (0.5)	Two sessions late (1)	One session late (1.5)	Early or ontime (2)
Efforts(4)	N/A	N/A	Not Completed (1)	Partially Completed (2)	Completed(4)
Legibility(2)	N/A	N/A	Poor(1)	Good(1.5)	Very Good(2)
Oral Assessment (2)	N/A	N/A	N/A	Partially Understood the concept (1)	Understood the concept(2)

```
from sklearn.datasets import load_iris
from sklearn.cluster import AgglomerativeClustering
import numpy as np
import matplotlib.pyplot as plt
from scipy.cluster.hierarchy import dendrogram ,linkage
```

```
data = load_iris()
df = data.data
df = df[:,1:3]
Z = linkage(df, method = "ward")
plt.figure(figsize=(30,30))
dendro = dendrogram(Z)
plt.title("Dendrogram")
plt.ylabel("Euclidean distance")
```

```
Text(0, 0.5, 'Euclidean distance')
```



9913 Postlab ML Exp6
Mark Lopes Batch-C

Q. What is Similarity based clustering
→ It is a technique in which data points are grouped into clusters based on how similar they are to each other. It uses measures such as Euclidean distance, cosine similarity or Jaccard ~~distance~~ index to determine closeness between objects. Points with high similarity are placed in same cluster, while dissimilar points fall into different cluster.

Q. Why do we need to perform significance testing in clustering.
→ It is important to ensure that the identified clusters represent genuine pattern in data rather than random groupings. Since clustering is unsupervised and lacks predefined labels, there is a risk of interpreting noise as structure. By performing significance testing, we can validate the quality and distinctiveness of clusters.

Q. Give example of confusion clustering to solve real-life problem.
→ It can be applied in healthcare for disease diagnosis. Patients symptoms and medical test results may overlap across different disease diseases creating confusion in classification. By using confusion clustering, patients of same patterns can be grouped together.

Q. difference between hard and soft clustering
→ • Hard clustering: Each data point has exactly one cluster. No overlaps, once a point is in a cluster, it cannot belong to another.
• Soft clustering: Each data point ~~&~~ can belong to multiple clusters with a certain probability or degree of membership.

Q. What are challenges associated with clustering
→ There are multiple challenges:-
1. Choosing no. of clusters in advance
2. Selecting an appropriate similarity measures.
3. Handling high-dimensional data.
4. Dealing with noise and outliers
5. Scalability to large datasets.

Q.